

Program: Diploma in Automation and Robotics	
Course Code: 5331	Course Title: Industrial Automation Control
Semester: 5	Credits: 4
Course Category: Program core	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To introduce the concepts of Process and Process control.
- To illustrates various elements in a process control loop and controller tuning.
- Familiarize the basic concepts of PLC programming
- To explain functional elements of process control loop and various communication methods in process control applications.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basics of electrical networks		Fundamentals of Electrical circuits and machines	2
Basics of Computers and programming		Introduction to IT systemsLab,, Python programming	1,3

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Understand the concept of Process control and the process control loops	15	Understanding
CO2	To understand the various control modes and to implement pneumatic and electronic controllers	16	Applying
CO3	Familiarize the basic concepts of PLC programming	16	Understanding
CO4	To understand telemetry system and Digital Communication channels in Process Control	13	Applying
	Series Test	2	

CO – PO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3						
CO2	3	2					
CO3			3				
CO4	3						

3-Strongly mapped

2-Moderately mapped

1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand the concept of Process control and the process control loops		
M1.01	Define the concepts of process control.	3	Understanding
M1.02	Explain automatic control	3	Understanding
M1.03	Explain the process and controller Characteristics	4	Understanding
M1.04	Describe the different process control loops	5	Understanding
Contents: Process- Process Control and Process Plant -Human aided control and Automatic control-Block diagram of process control - Error, set point, controlled variable, manipulated variable and measured variable-Elements of process control loop-Process characteristics - Process equation, Process Load, Process Lag and Self regulation, Control System Parameters - Error, Variable Range, Control Parameter Range, Control Lag –Dead Time, Cycling-Process Parameters-Temperature process control, Pressure Process control system-Flow process control system, Level process control system.			
CO2	To understand the various control modes and to implement pneumatic and electronic controllers		
M2.01	Describe different types of process	4	Understanding
M2.02	To understand the various control modes continuous and discontinuous	6	Understanding

M2.03	Understand the implementation of Electronic Controllers	3	Understanding
M2.04	Illustrate Tuning of controllers	3	Applying
	Series Test – I	1	
Contents: Batch and Continuous process - Discontinuous control Modes - two position - neutralzone -multi position control Mode- Continuous Control Modes - proportional controlMode -proportional band and offset error - integral control Mode – reset rate – derivative control Mode – derivative time- composite control Modes- PI, PD and PID control Modes–Electronic Controllers-Electronic Error detector-Electronic P,PI,PD and PID controller using Op -amp-Tuning methods– open loop method-Ziegler Nichols method.			
CO3	Familiarize basic concept of PLC programming		
M3.01	Understand Sequence and logic control	1	Understanding
M3.02	Describe a programmable logic controller and its components	2	Understanding
M3.03	Understand Ladder diagram	3	Applying
M3.04	PLC programming	4	Applying
M3.05	Case studies on PLC Ladder programming	3	
M3.06	Understand various Motion Controllers	3	
Contents: Sequence control and programmable controllers – logic control and sequencing elements, ladder diagram, PLC, programming of PLC- analog and digital I/Os, timers, counters, function blocks. Case studies on PLC ladder programming. Motion controllers-VFD, MLD, external relays and contactors.			
CO4	To understand telemetry system and Digital Communication channels in Process Control		
M4.01	Describe the different telemetry systems	3	Understanding
M4.02	Explain the concept of foundation Field bus and Profit bus DP	6	
M4.03	Explain the concept of HART Communication Protocol	3	Applying
	Series Test – II	1	
Contents: Telemetry system- General Telemetry system blockDiagram- Voltage Telemetry System- Current Telemetry System - Motion balance andForce balance telemetry systems - Position Telemetry System- Digital communicationchannels-Fieldbus-advantagesoffieldbus-ProfibusDP-Functionalelements inFoundationFieldbus.Highway –			

Addressable Remote Transducer-Benefit of HART field communication protocol. Working of HART - Block diagram of HART digital communication System – HART Specification (Layers of HART).

Text / Reference

T/R	Book Title/Author
T1	Industrial Instrumentation, Control and Automation, S. Mukhopadhyay, S. Sen and A. K. Deb, Jaico Publishing House, 2013
R1	Jonathan Love, Process Automation Handbook, Springer 2007
R2	Curtis.D. Johnson, Process control instrumentation technology, PHI.
R3	Liptak, Volume II, Instrument Engineers Hand book, Chilton Book Company.
R4	Krishnakanth, Computer based Industrial Control, PHI.
R5	Donald.P. Eckman, Automatic process control, WILEY India.

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/108105088
2	https://onlinecourses.nptel.ac.in/noc21_me67/preview
3	https://scienceq.org/wp-content/uploads/2018/10/JAETV7I102.pdf
4	https://processcontrol.tv/tutorials
5	https://www.pidlab.com/en/virtual-labs